Lab Sheet 03 · Part C Answers

HDIT 12103 Website Design and Development · Mini project: Sign-up form

Files

`signup.html` goes in the project root. `signup.js` goes in the `js` folder.

Step 2: The five TODOs

TODO 1: checkUsername

```
function checkUsername(value) {
  if (!/^\w{3,15}$/.test(value)) {
    return "Use 3 to 15 letters, numbers or _.";
  }
  if (takenUsernames.includes(value.toLowerCase())) {
    return "That username is taken. Try another.";
  }
  return "";
}
```

TODO 2: checkEmail

```
function checkEmail(value) {
  if (!/^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(value)) {
    return "Enter an email like name@example.com.";
  }
  return "";
}
```

TODO 3: checkPasswordRules

```
function checkPasswordRules(value) {
  const results = {
    length: value.length >= 8,
    number: /\d/.test(value),
    upper: /[A-Z]/.test(value),
    symbol: /[^A-Za-z0-9]/.test(value)
  };
  let metCount = 0;
  for (const key in results) {
    ruleItems[key].classList.toggle("met", results[key]);
    if (results[key]) metCount++;
  }
  showStrength(value, metCount);   // Challenge 1
  return metCount === 4;
}
```

TODO 4: checkConfirm

```
function checkConfirm() {
  if (confirmBox.value === "") return "Please type your password again.";
  if (confirmBox.value !== password.value) return "Passwords do not match.";
  return "";
}
```

TODO 5: Show / Hide button

```
showBtn.addEventListener("click", () => {
  const hidden = password.type === "password";
  password.type = hidden ? "text" : "password";
  showBtn.textContent = hidden ? "Hide" : "Show";
});
```

Step 3: Test table

Test	Expected result	Result
Open the page	Create account is disabled; all four rules show X in grey	Pass

Test	Expected result	Result
Username: Muzawir	That username is taken. Try another.	Pass
Username: ab	Use 3 to 15 letters, numbers or _.	Pass
Username: muzawir_r	No error	Pass
Email: rizwanmuzawir@gmail	Enter an email like name@example.com.	Pass
Password: icst2026	Two rules ticked: 8 characters and a number	Pass
Password: Icst2026!	All four rules ticked	Pass
Confirm: Icst2026	Passwords do not match.	Pass
Click Show, then Hide	The password appears as text, then as dots again	Pass
Confirm Icst2026! and tick the box	Create account becomes enabled	Pass
Click Create account	Welcome, muzawir_r! Your account is ready.; the form disappears; the password is not shown anywhere	Pass

"Muzawir" is in the pretend taken-username list (the `takenUsernames` array in the code), so it shows the taken message, and capitals are ignored. My own username for the passing tests is **muzawir_r** and my email is **rizwanmuzawir@gmail.com**. The results above were checked by running the code against these steps. Repeat them in your own browser and tick each row.

Full sign-up run with my own details

Field	What I typed	Result
Username	muzawir_r	No error (9 characters, letters and _ only, not in the taken list)
Email	rizwanmuzawir@gmail	Enter an email like name@example.com. (no dot after the @)
Email (corrected)	rizwanmuzawir@gmail.com	No error
Date of birth	left empty (optional)	No error; a date under 16 years ago would show the age error
Password	Icst2026! (the sheet's test password)	All four rules ticked; Strength: Strong
Confirm password	Icst2026!	No error (matches)
Campus IT rules box	ticked	Create account becomes enabled
Click Create account		Welcome, muzawir_r! Your account is ready.; the form disappears

Refresh the page after this run: the username muzawir_r is filled in again, because Challenge 4 remembers it in localStorage (the password is never stored). The Start again button clears it.

Task C: Challenge tasks (all five done)

#	Challenge	How it works
1	Strength word	`showStrength()` counts the rules met: Weak (0 to 2), Fair (3), Strong (4), shown under the checklist.
2	Caps Lock warning	A `keydown` listener on the password boxes uses `e.getModifierState("CapsLock")` and shows "Caps Lock is on."; it clears on blur.

#	Challenge	How it works
3	Date of birth 16+	`checkDob()` works out the age from the date and gives an error for anyone under 16 or a future date. The field is optional, so the test table above still passes.
4	Remember username	Only the username is saved in `localStorage` (key `signup-username`) and filled in on reload. The password is never stored.
5	Start again button	Resets the form, clears every error message, unticks all rules, clears the strength word and the remembered username, and shows the form again.

Security note: passwords

The code never shows the password back to the user, never calls `console.log` on it, and never saves it in `localStorage`. Only the username is stored. The `takenUsernames` list is pretend: a real site must ask the server, because only the server knows every account.

Commit message

Complete Part C mini project